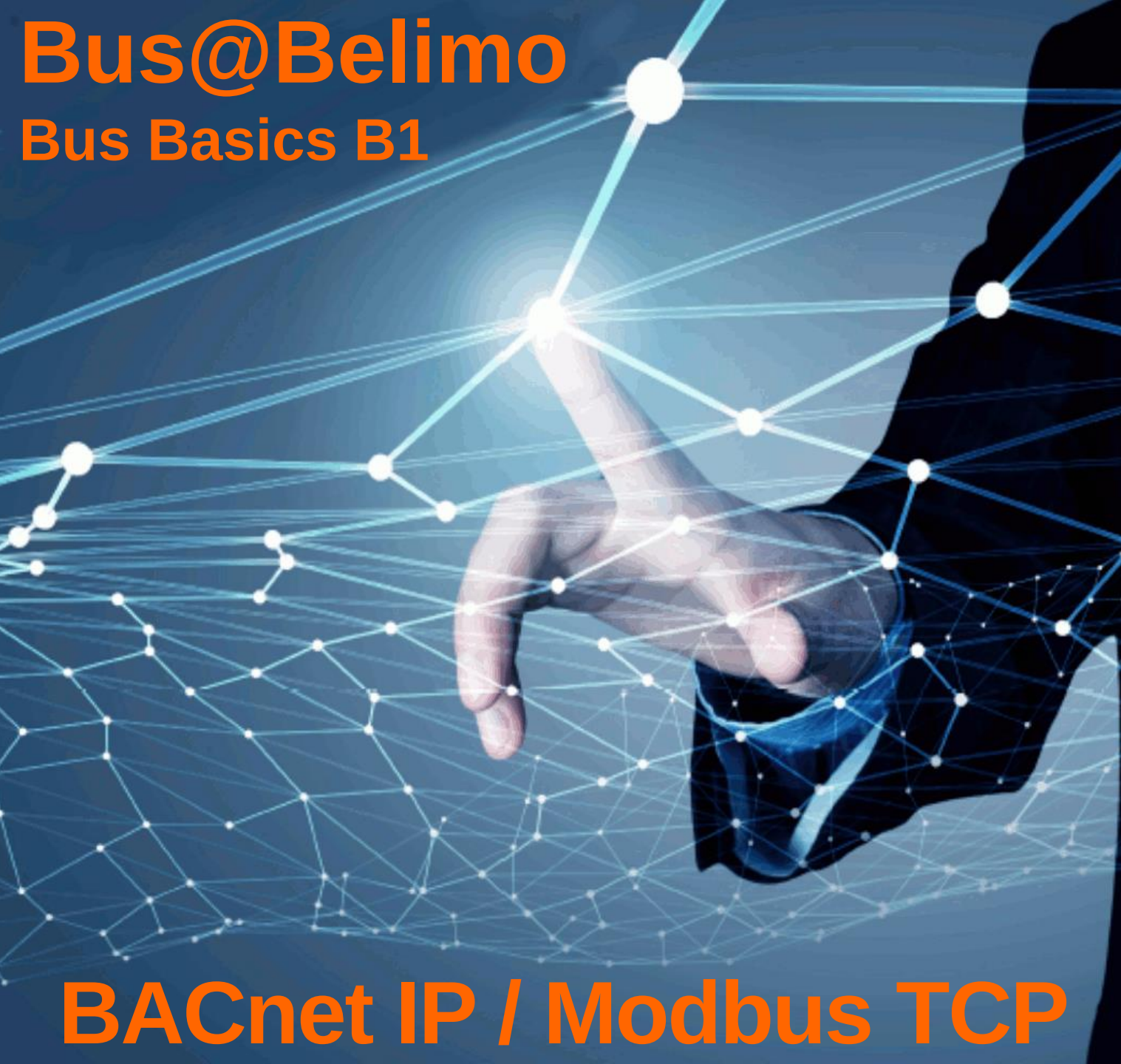




Bus@Belimo

Bus Basics B1



BACnet IP / Modbus TCP



Agenda Wednesday



BACnet MS/TP

BACnet IP / Modbus TCP

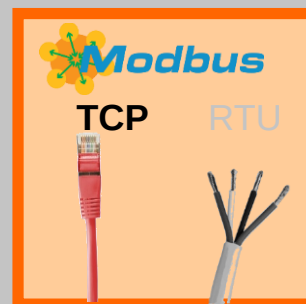
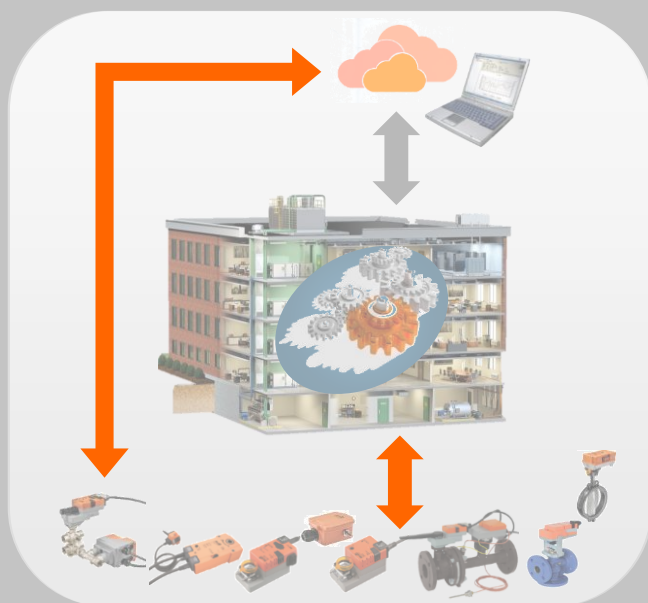
Lunch@Belgusto

MP-Bus

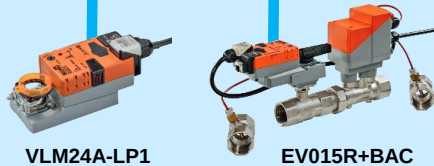
MP-Bus

Dinner together

Modbus@Belimo BACnet@Belimo



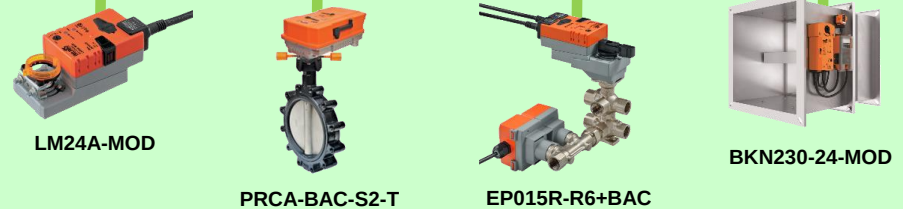
Modbus IP and Modbus MS/TP



VLM24A-LP1

EV015R+BAC

Modbus TCP:
IP Ethernet Communication



LM24A-MOD

PRCA-BAC-S2-T

EP015R-R6+BAC

BKN230-24-MOD

Modbus RTU:
RS-485 communication

Modbus TCP Addressing



IP Subnet: 192.168.0.1...254
Subnet Mask: 255.255.255.0

**Modbus
MASTER**



Port: 502, Timeout = 30s

Addr: 1



VLM24A-LP1

Addr: 2



EV015R+BAC

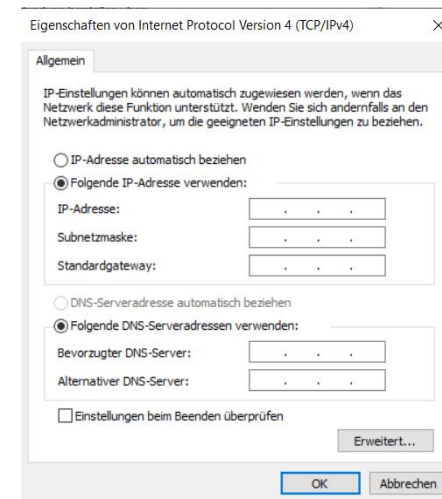
Parameterisation



IP Configuration for Belimo Devices

IP Address	For example: 192.168.1.10	IP Address , corresponding to the IP Subnet (IP V4 address)
Subnet Mask	For example: 255.255.255.0	Subnet Mask of the IP Ethernet network (Definition of Subnet Size)
Default Gateway	For example: 192.168.1.50	Default Gateway of the IP Ethernet subnet (Address to which all packets with IP addresses from other subnets should be sent: IP Router to other networks)

➔ *IP Addressing must be configured according to the specific IT Network*



Modbus TCP - Configuration



Modbus TCP Configuration for Belimo Devices

Modbus Address	1-247	Modbus Address, is only used for TCP in case that the Modbus device is connected to a Modbus RTU network. The IP Address for TCP Communication is configured in the IP settings of the device.
Port	Default: 502	Can be selected free. Default is 502.
Timeout	1-99	Seconds the actuator keeps the connection open. Default 30s



Communication protocol

- ☐ BACnet IP
- ☐ BACnet MS/TP
- ☐ MP
- ☒ Modbus TCP
- ☐ Modbus RTU
- ☐ None

Modbus TCP settings

Modbus address

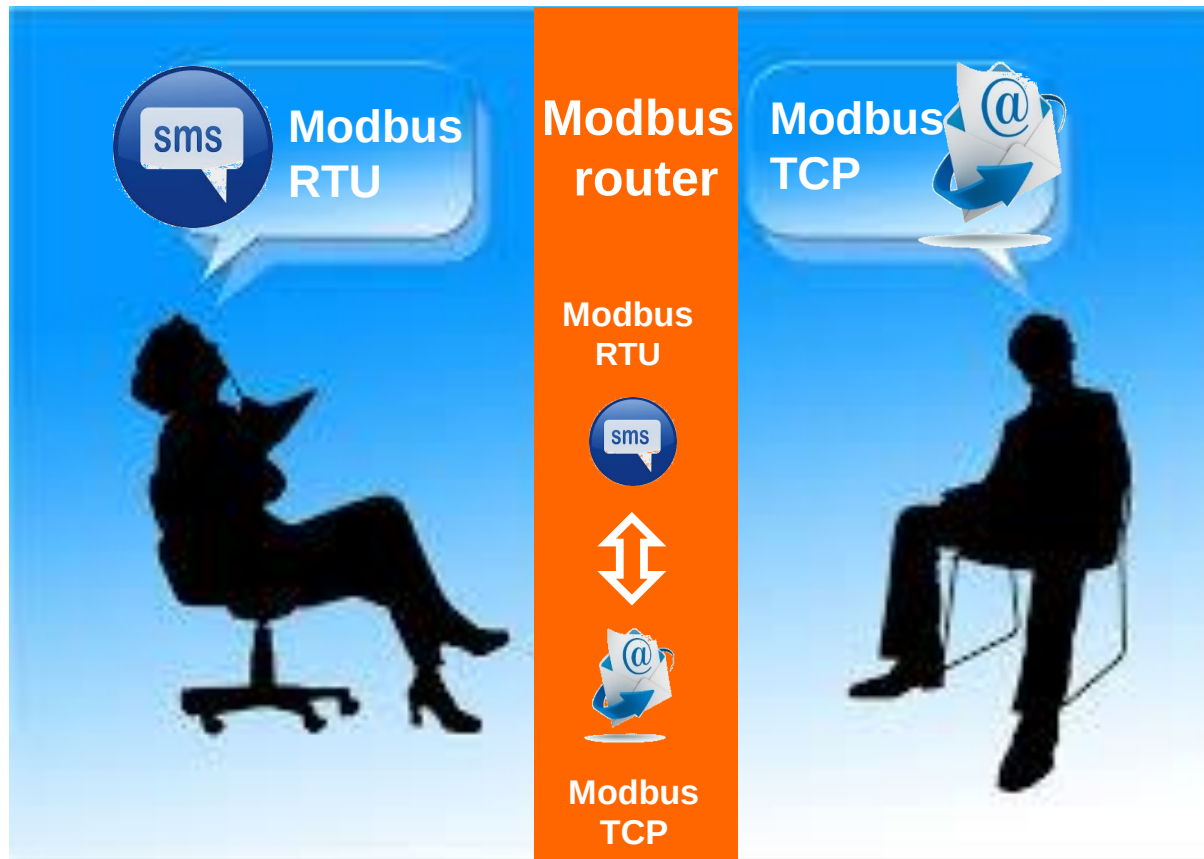
TCP port

Keep open timeout [seconds]

Transport / Transmission: Router



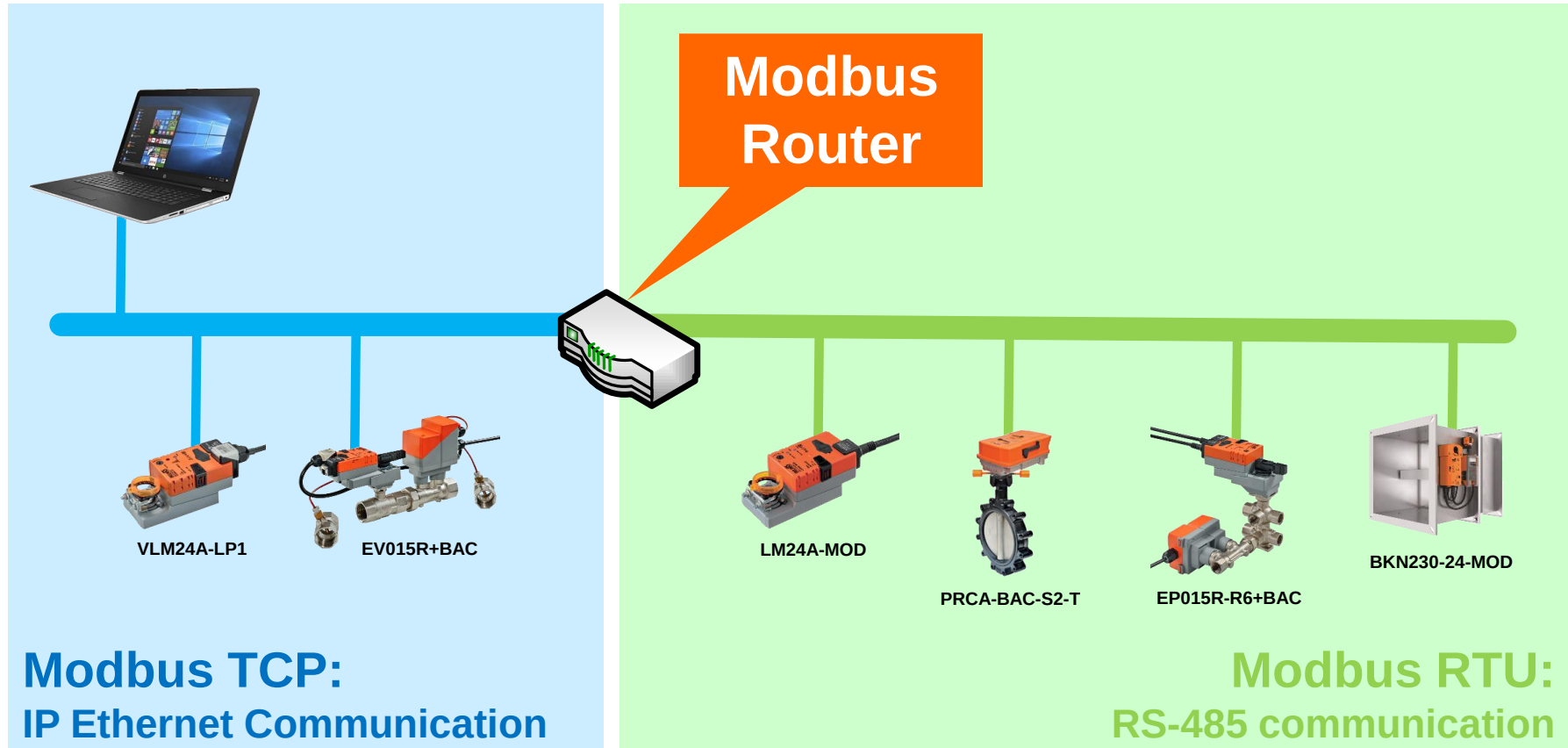
Connection of different Transport Options



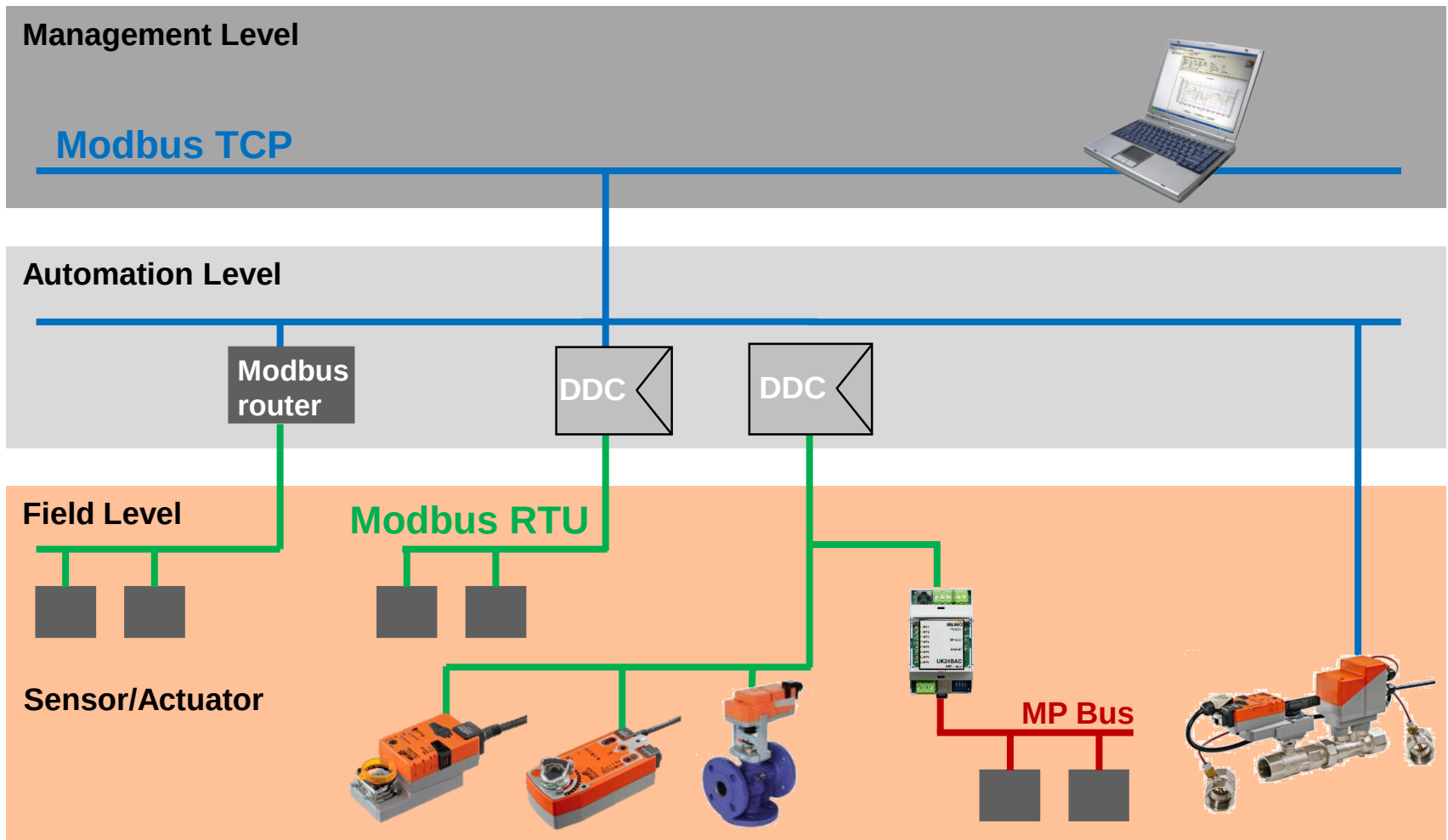
Modbus TCP and Modbus RTU



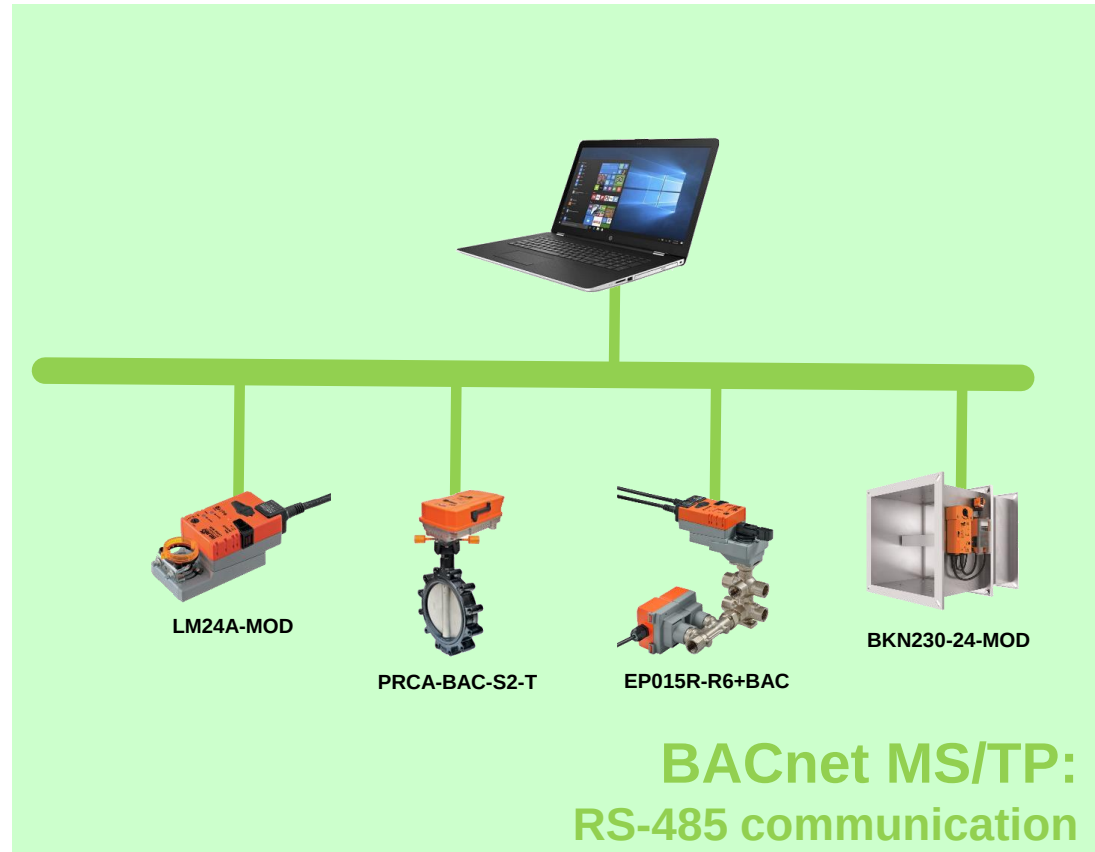
➔ *Modbus addresses must be unique in the entire Modbus line!*



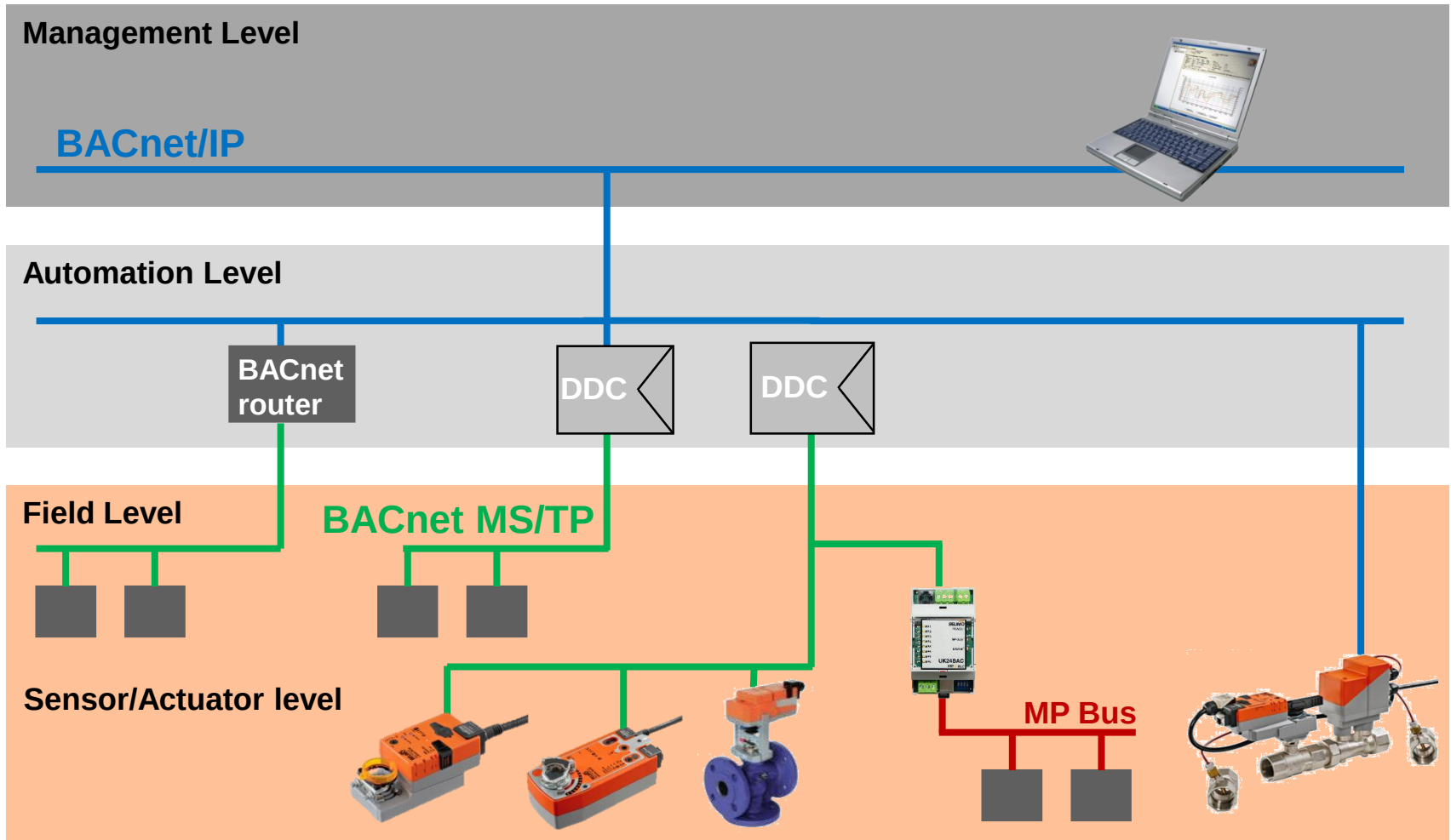
Modbus TCP and Modbus RTU



BACnet IP and BACnet MS/TP



BACnet IP and BACnet MS/TP



BACnet IP Addressing



IP Subnet: 192.168.0.1...254
Subnet Mask: 255.255.255.0

InstNr: 10

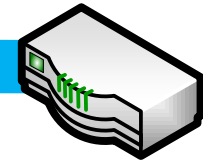
IP Addr: 192.168.0.10



BACnet Client - Yabe

InstNr: 1000

IP Addr: 192.168.0.100



BACnet Router

UDP Port: 0xBAC0 = 47'808

InstNr: 2

VLM24A-LP1

IP Addr: 192.168.0.2



InstNr: 3

EV015R+BAC

IP Addr: 192.168.0.3



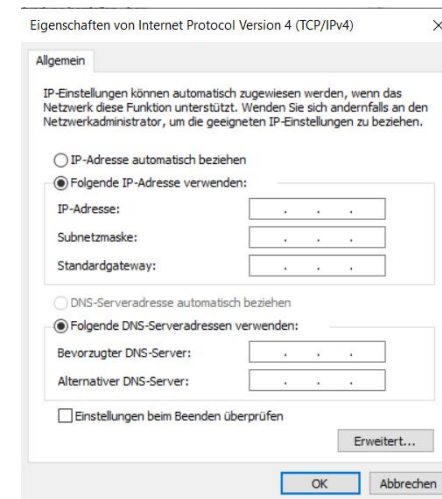
Parameterisation



IP Configuration for Belimo Devices

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➔ *IP Addressing must be configured according to the specific IT Network*



BACnet IP Configuration for Belimo Devices

Device Name	Site4'Floor2 'Zone3'VAV1	BACnet Internetwork-wide unique Text for the Device Identification
Instance Number	1 - 4194302	BACnet Internetwork-wide unique Number for the Device Identification
Port	Default: 47'808	Can be selected free. Default is 47'808 (0xBAC0).



Communication protocol

- ☒ BACnet IP
- ☐ BACnet MS/TP
- ☐ MP
- ☐ Modbus TCP
- ☐ Modbus RTU
- ☐ None

BACnet IP settings

47808 Port

☒ Simple device

☐ Foreign device

Device object settings

1210 Instance ID

EV3 Demo Device name

Energy Valve Device description

AP Location

0 System status

1 Protocol version

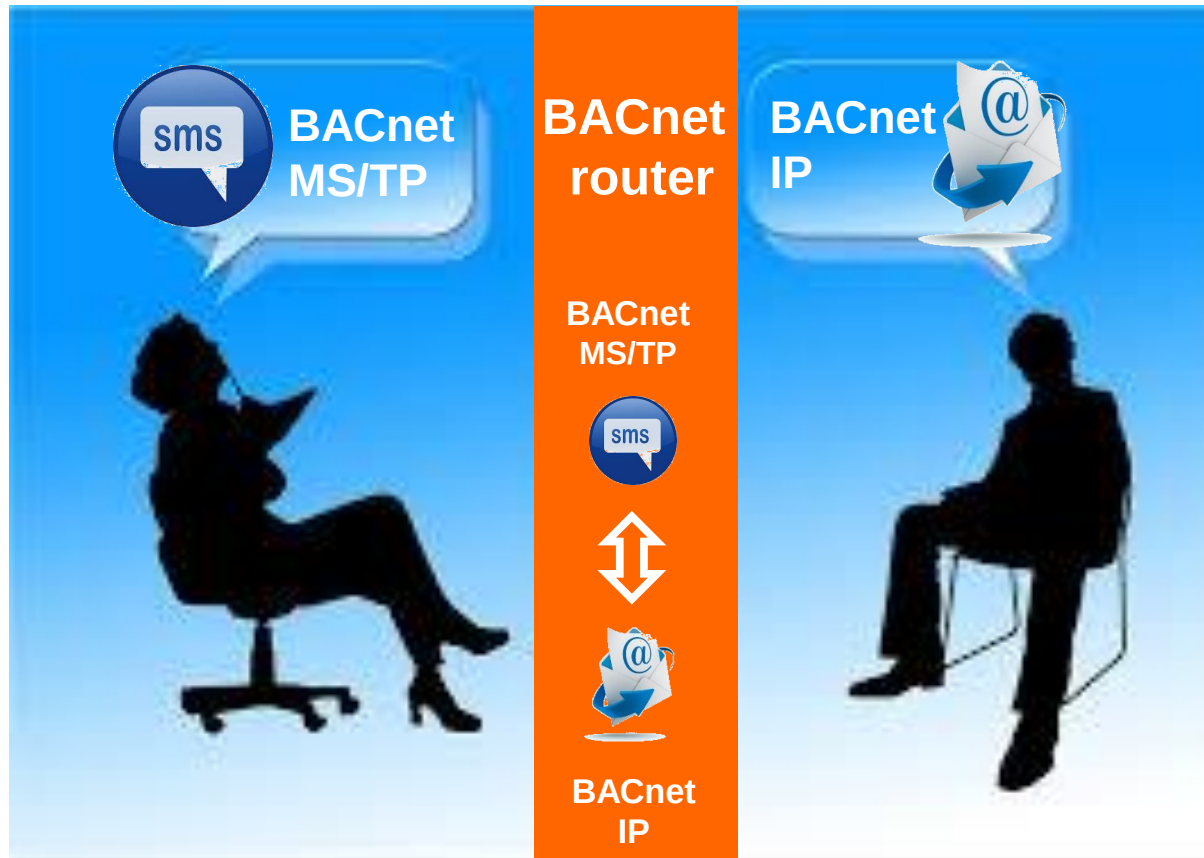
12 Protocol revision



BACnet Instance Number and Device Name must be unique in the entire BACnet Internetwork!

Transport / Transmission: Router

Connection of different Transport Options



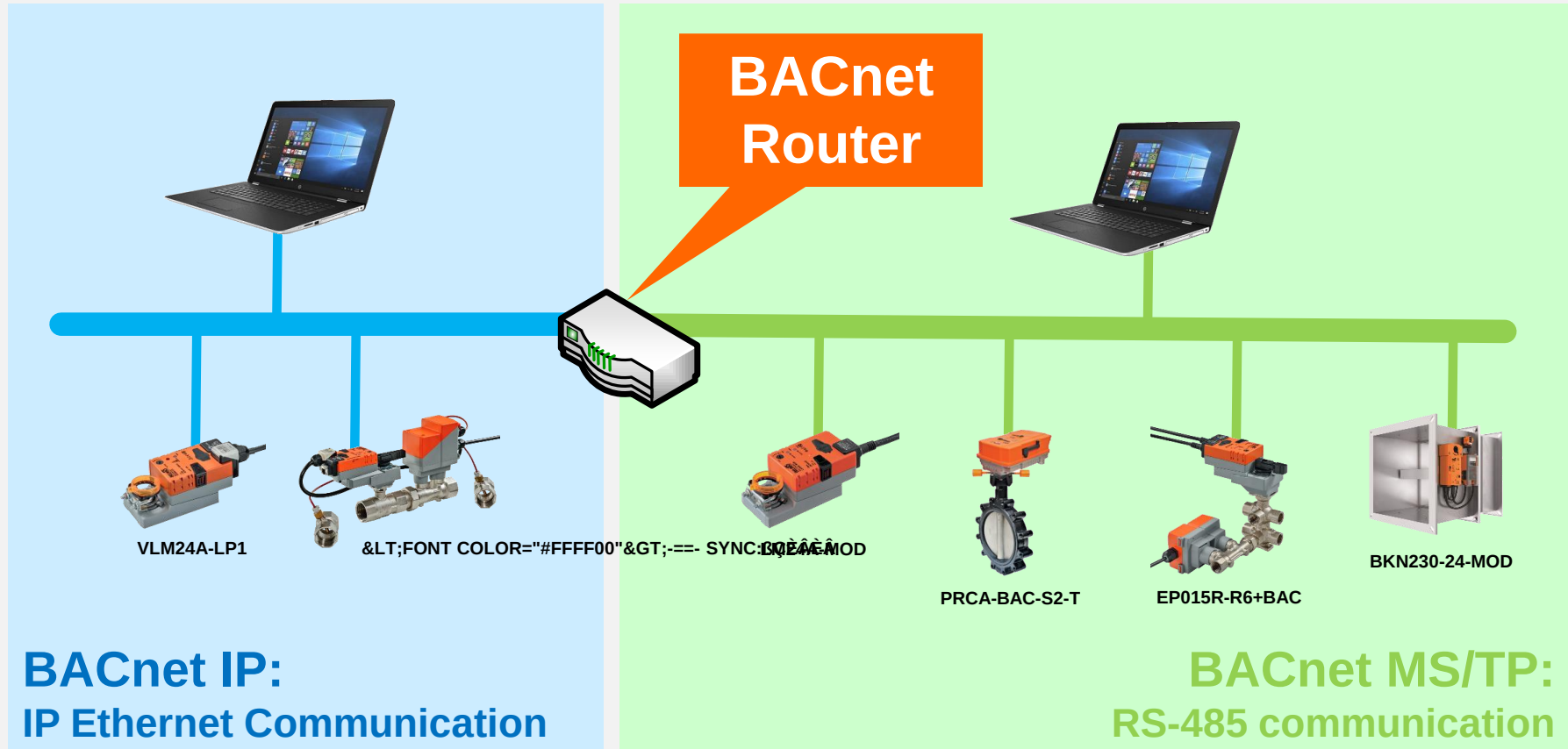
BACnet IP and BACnet MS/TP



BACnet Internetwork



BACnet Instance Number & Device Name must be unique in the entire BACnet Internet network!





BELIMO®
